20.2. The amount of steam found in example 20.1 is released into an air-filled large dry containment where the pressure may not exceed 0.7 MPa. Assume steam and air in the containment can be treated as ideal gases. Estimate the necessary containment volume.

20.3. In the Chernobyl accident, reactor output increased from ~20% to ~100 times full power in about 4 seconds. Assume a constant reactivity excess during this time. a) What was the average reactor period? b) How much energy was released during this time?

20.4. KOH is normally used for pH control in VVER reactors. Which radioisotopes will be formed from potassium in the coolant?

20.5. ^{232}U is formed in a thermal ^{232}Th based breeder. a) By which chain of reactions is it formed? b) Will the choice between a continuous or discontinuous (once a year) reprocessing of the Th blanket affect the isotopic composition of the isolated uranium (assume that all other actinides remain in the blanket)?

20.6. Use eqn. (19.43) to estimate the maximum thermal efficiency for the Creys-Malville plant.

20.7. Assume that the probability of not restoring AC power within 20 minutes is reduced by a factor of 5 in Fig. 20.9. a) Which is now the risk dominating event chain? b) Is it better to reduce the total risk in Fig. 20.9 by this change or by improving the stability of the HV mains with a factor of 2 by e.g. building more power stations?

20.8. What is the fate of the gas released into the turbine condensers during regeneration of the charcoal absorbers in Fig. 20.6?

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CHAPTER 21

The Nuclear Fuel Cycle

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